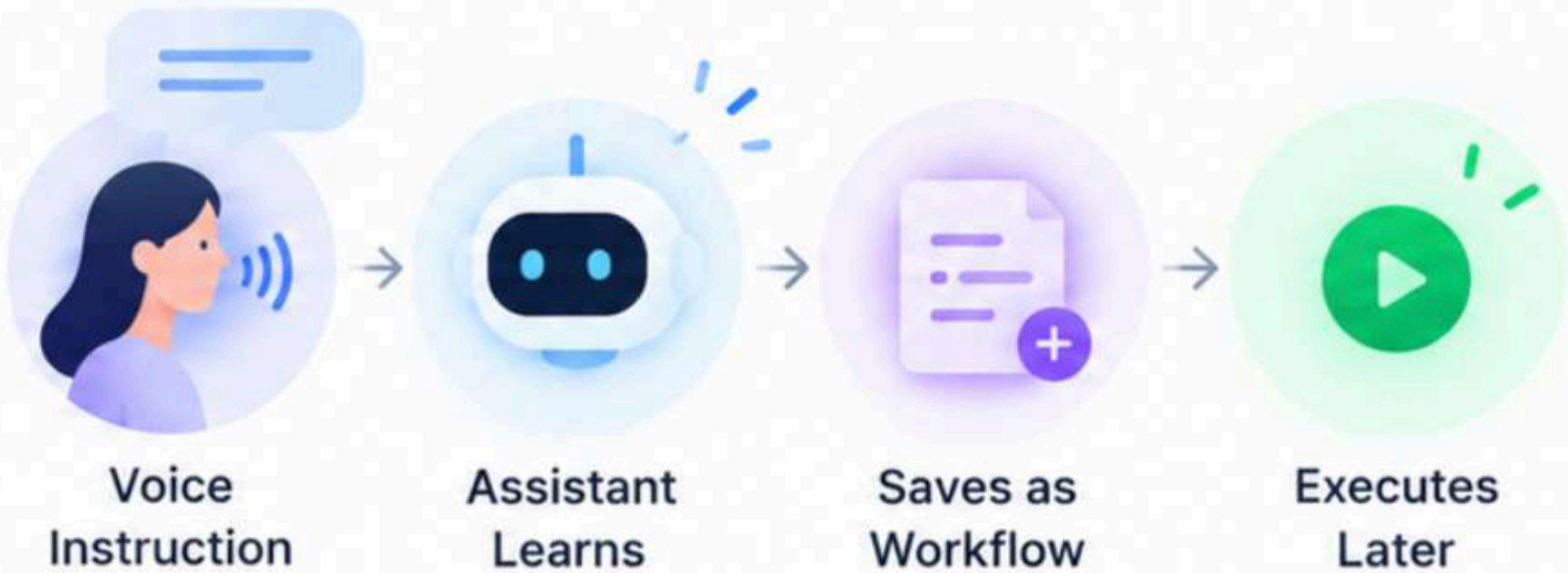


Generative AI Hackathon

3rd Edition 2026 – 27

Teachable Voice Automation

Teach an assistant a task once using natural voice instructions.



THEME ID

TEACHABLE VOICE AUTOMATION

TEAM NAME

CHOCOCHIPCOOKIES

COLLEGE NAME

MS RAMAIAH INSTITUTE OF TECHNOLOGY

TEAM MEMBERS

Member 1

PRIYAL LAKHERA

Member 3

SUDHANSHU KUMAR

Member 2

ADITI PRIYA

Member 4

SHANU KESHRI

SUBMISSION LINK

 [github.com/repo-link](https://github.com/priyallakhera01/TAV.git)

<https://github.com/priyallakhera01/TAV.git>

 [Demo Video Link](#)

https://drive.google.com/file/d/1lxvRJ4u7mLaf9M77QkqxQz_EHH29XfEC/view?usp=drivesdk

Theme: Teachable Voice Automation

The Problem Scenario

Users know what they want to do and the steps to do it, but there is no simple way to teach these steps to the assistant. For users with motor disabilities, limited dexterity, tremors, visual impairments, temporary injuries, or when both hands are occupied, manually configuring multi-step app workflows is difficult and frustrating.

Our Goal

Enable users to teach a task once using natural voice instructions. The assistant learns the demonstrated workflow and can execute it later from a simple voice command.

Execution Pipeline



Example Request

-  "Open the cab app, set my usual pickup location, and book a ride."
-  "Every morning, open my calendar and read my first meeting."
-  "When I receive a bill, remind me before the due date."
-  "Turn on the lights and start my study playlist."

Target Audience

-  Motor disabilities & limited dexterity
-  Tremors / temporary injuries
-  Visual impairments
-  Hands-occupied situations (driving, cooking)

Existing Solutions & Gaps

UI Automator & Appium

Require manual script writing and hardcoded locators/coordinates. Not designed for natural, voice-based task teaching.

Tasker + AutoInput

Consumer macro tool. Relies on fixed coordinate replay and manual rule configuration. No understanding of the intended workflow.

RPA Systems

Built for enterprise use with predefined rules. Require complex setup and manual workflow construction. Not accessible for everyday users.

LLM / GUI Agents

Heavy cloud models or complex GUI agents. Often lack reliable grounding for real apps, are resource-intensive for on-device use, and do not support reusable, personalized workflows.

THE EXISTING PARADIGM (BROKEN)

Users are forced to manually configure rules, write scripts, or rely on rigid X/Y coordinate macros that break with any UI change.

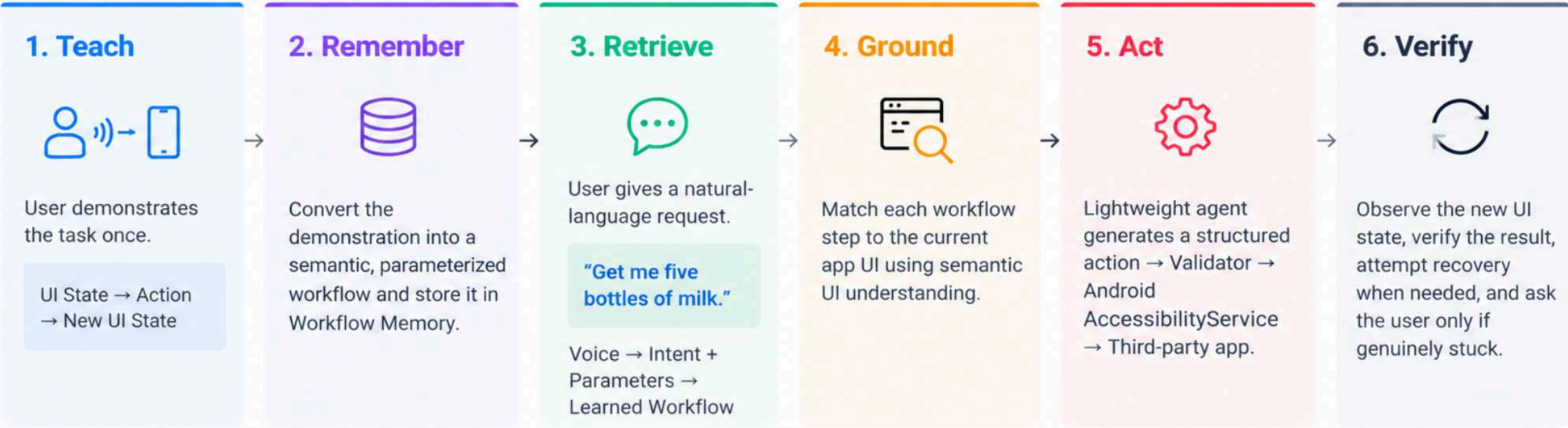
OUR TEACHABLE PARADIGM

Teach a task once → Using **natural voice** instructions → System understands and structures the workflow → Saves it for real **use and executes later.**

Our Solution

Teach Once → Remember → Retrieve → Ground → Act → Verify

A voice-driven, teachable automation system that converts a user's one-time demonstration into a reusable semantic workflow for third-party Android apps.



Teach once. Speak naturally. Let the assistant execute the workflow.

Our Solution & Architecture



Workflow Memory
Learned workflows, parameters, examples, traces

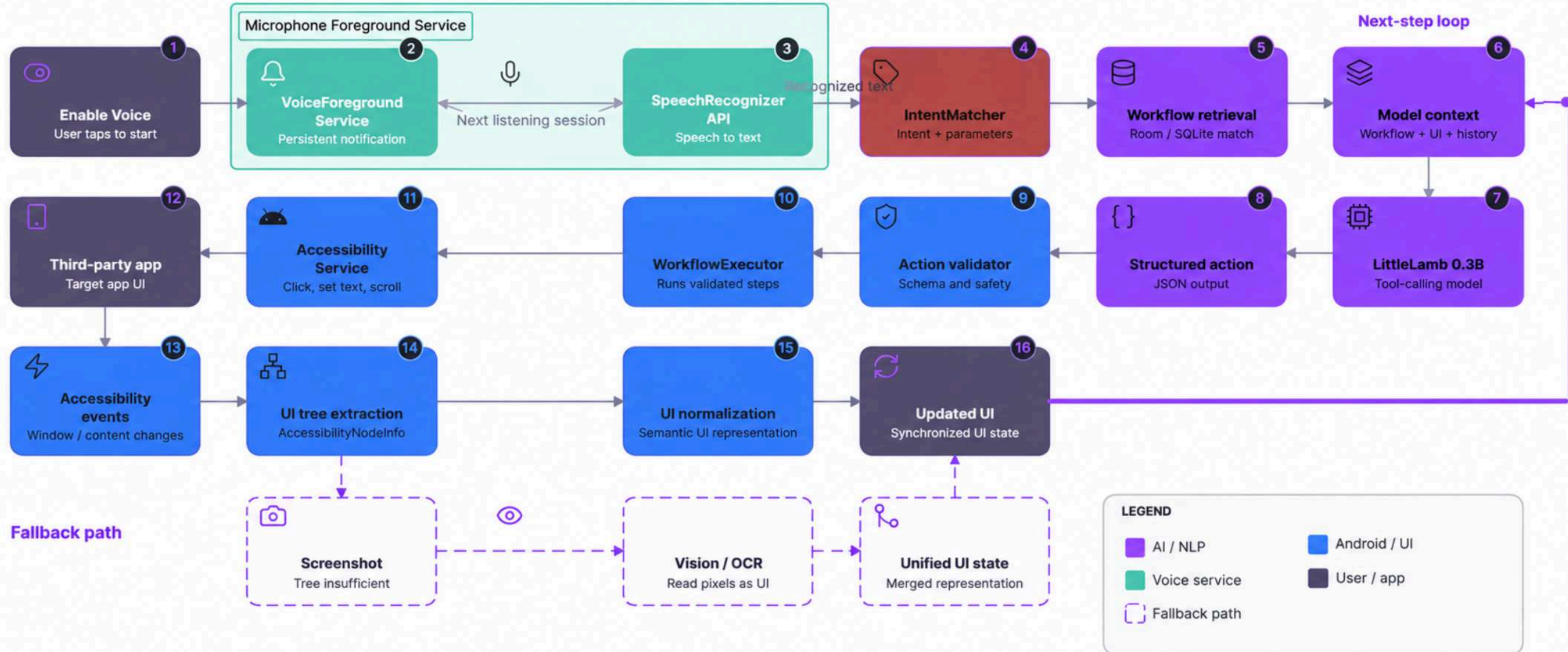
UI Understanding
Extracts and normalizes accessibility trees

Model Brain
Selects the next structured action

Action Executor
Maps actions to AccessibilityService

Verification
Confirms the intended effect occurred

Voice-Controlled Android Automation Agent



Demo & Product Walkthrough



A. Teach Once

"Teach me how to order a paneer pizza."

User performs task: Search → Select → Qty → Cart.

System records: UI State → Action → New UI State.



B. Workflow Synthesis

Raw coordinates converted to semantic actions.

Instead of: Tap(183,542)

Store: CLICK(search_field)
INPUT(product)



C. Voice Request Extraction

"Get me a large paneer pizza."

```
{  
  "intent": "buy_food",  
  "item": "paneer pizza",  
  "size": "large"  
}
```



D & E. Retrieval & Replay

Retrieval: NLP matches intent to learned workflow, seamlessly handling paraphrases.

Replay: Find semantic target → Execute → Observe new UI → Verify.



F. Sensitive Boundary

System pauses for security on:

- PIN / OTP
- Passwords / CVV
- Payment Confirmation



G. Error Recovery

If an element is missing or UI state is completely unexpected:

Re-observe → Attempt Recovery → Ask user only if genuinely stuck.

Tools and Tech Stack Used



Android Base

Kotlin / Android API

AccessibilityService events

AccessibilityNodeInfo trees

ACTION_CLICK, SET_TEXT

Focus, Scroll, Back gestures



UI Understanding

Tree Extraction logic

UI Normalization pipeline

Semantic UI representation

UI State Synchronization

Fallback OCR (Planned)



Workflow Layer

Demonstration Recorder

Workflow Synthesis engine

Room / SQLite memory

Workflow Parameters store

Execution History tracking



AI & NLP

Speech-to-Text module

Intent Extraction logic

Semantic Retrieval matching

LittleLamb 0.3B tool-calling

Structured JSON actions

IMPACT AND USE CASE



Motor & Dexterity Limits



Visual Impairments



Hands-Occupied (Situational)

How it exactly helps:

- ✓ Reduces repeated touch interactions
- ✓ Enables hands-free task setup

How it exactly helps:

- ✓ Reduces dependence on navigating complex interfaces
- ✓ Allows tasks to be taught conversationally

How it exactly helps:

- ✓ Allows workflows to be taught without navigating menus
- ✓ Helps get everyday tasks done quickly and safely



Cooking

"Start a 15-minute timer and play my cooking playlist."



Travel

"Book my usual cab to college."



Routine

"Remind me to take my medicine after dinner."



Study

"Open my notes, start a 50-minute timer and turn on Do Not Disturb."



Home

"Turn on the lights and play my evening playlist."



Bills

"Remind me three days before my electricity bill is due."

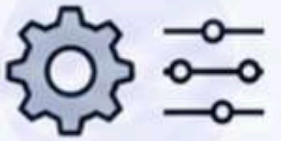
Innovation Highlights & Limitations

Innovations



Teach by speaking (Accessible automation)

Users describe the task naturally in voice. Automation becomes teachable — not just configurable through interfaces.



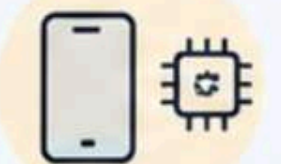
From rule-based to semantic workflows

Demo trajectory immediately becomes a structured workflow with semantic understanding (intent, steps and parameters).



Disability-inclusive automation

Voice-first task teaching reduces dependence on precise touch interaction, making multi-step workflows more accessible for users with motor disabilities, tremors, and visual impairments.



On-device agent

Lightweight model (LittleLamb 0.3B) runs locally with zero cloud dependency and low latency.



Closed-loop execution

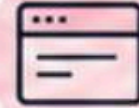
Observe → Decide → Act → Verify ensures reliable task execution.

Current Technical Limitations



App-level restrictions

Some applications do not expose all actions to automation frameworks (e.g., certain in-app functions or system-level settings).



UI variability

Changes in app layouts or UI elements can break coordinate/UI-based actions and require re-interpretation.



Permission requirements

Certain actions require Android accessibility permissions or additional system permissions, which may need explicit user approval.



Ambiguous instructions

Natural-language instructions may need clarification when the intended workflow is unclear or underspecified.



Execution reliability

External conditions such as network availability, login state or session expiry can affect execution.



LLM dependency

Semantic understanding may require a capable language model, while on-device execution must remain lightweight and resource-efficient.

Key Differentiation

Traditional Macros

- ✗ Records **fixed UI actions or coordinates**.
- ✗ Breaks when the **UI structure or layout changes**.
- ✗ Requires **manual configuration** for variations.
- ✗ Replays actions **without understanding user intent**.

Our Teachable Agent

- ✓ **Teach by speaking, not configuring rules.**
Users describe the task naturally; the agent converts it into a reusable workflow.
- ✓ **Learns semantic UI intent, not coordinates.**
- ✓ **Generalizes parameters from natural-language intent.**
- ✓ **Observe → Act → Verify closed-loop execution.**



Teach your phone a task once.

Later, describe what you want naturally—and let the agent **execute the learned workflow**.

Brownie Points

Small
extras.
Big impact.



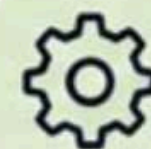
Personalized skills

Learns and remembers your frequently used tasks and adapts to your preferences.



Works across apps

Handles multiple third-party Android apps in a single workflow.



Parameter flexibility

Adapts to changing inputs like quantity, location, date or time without re-teaching the task.



More accessible

Voice-based teaching makes automation possible for users with limited dexterity or visual impairments.



Adaptive execution

Handles minor UI changes and recovers from failures by re-observing the screen and adjusting.



No manual rules

No need to configure triggers, conditions or actions — just teach it once through natural interaction.

→ Everyday automation, now simpler, smarter and more inclusive.

What's Next

1. Vision / OCR Fallback

If the Accessibility Tree is insufficient or intentionally obfuscated
→ Capture Screenshot → Run on-device Vision/OCR →
Synthesize a Unified UI State.

2. Better Workflow Retrieval

Move from basic matching to Embedding Retrieval → Top-K
Candidates → Confidence Reranking before execution to handle
highly similar workflows flawlessly.

3. Continuous Personalization

Successful executions automatically append the user's natural-
language invocation back into memory, continuously adapting to
their unique vocabulary and slang over time.

4. Expanded Coverage & Reliability

Scale beyond initial test apps (Amazon, Zomato). Rigorously
improve state verification logic and refine user escalation
triggers so the agent only asks for help when truly necessary.

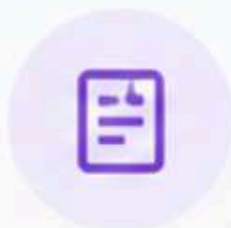
Submission Checklist



Working Prototype Code

Complete, runnable code for core features.

✓ Yes



Reproducible Setup README

Clear setup instructions to reproduce the demo.

✓ Yes



Demo Video (Max 5 mins)

Covers key features and example workflows.

✓ Yes

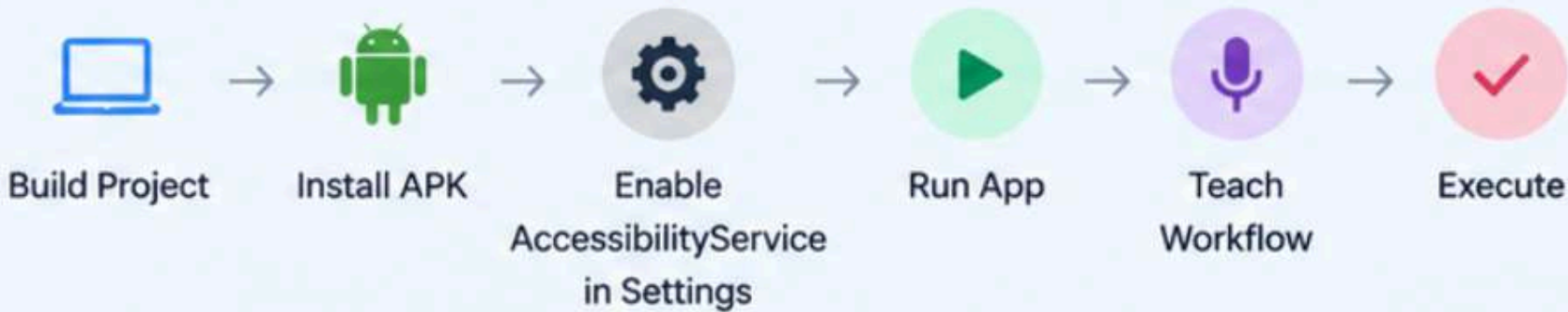


Presentation File

Final deck summarizing problem, approach, demo and results.

✓ Yes

REPRODUCIBLE SETUP



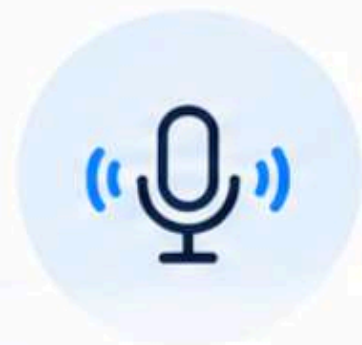
GITHUB REPOSITORY STRUCTURE

```
Android_Prototype/
├── AccessibilityService/      # UI interaction & automation
├── UI_Extraction_Normalizer/  # Screen parsing (Vision/OCR + AT)
├── Demonstration_Recorder/    # Demo recording & workflow generation
├── Workflow_Synthesis/        # Intent → steps → executable workflow
├── Intent_Vector_Retrieval/   # Workflow retrieval using embeddings
├── LittleLamb_Integration/    # On-device model integration
└── Verification_Recovery/     # Execution monitoring & error handling
```


Thank You!

Teach it once. Let your assistant do it **again.**

Teachable Voice Automation



Voice

You describe
the task naturally



Understand

Agent interprets
intent and steps



Learn

Saves as a reusable
workflow



Execute

Runs it for you
across apps

ORGANISED BY THE LANGUAGE AI TEAM AND THE PRISM TEAM

Samsung R&D Institute India